

The non-parametric/machine learning future?

Chris Conlon

Fall 2026

NYU Stern

Nonparametrics?

What do you mean by non-parametric?

Mostly we mean putting a flexible distribution on $f(\beta_i, \alpha_i \mid \theta)$ (and keeping logit error on ε_{ij})

$$u_{ij} = \beta_i x_j - \alpha_i p_j + \xi_j + \varepsilon_{ij} \text{ with } f(\beta_i, \alpha_i \mid \theta)$$

- Fixed Grids (Fox, Kim, Ryan, Bajari 2011, Heiss, Hetzenecker, Osterhaus 2022, Nevo Turner Williams 2016): draw from “prior” of β_i , compute $\sigma_{ij}(\beta_i)$ and choose weights on each i, w_i .
- Compiani (QE 2022): approximate $\sigma_j^{-1}(\mathbf{s}_t, \mathbf{x}_t^{(2)})$ directly with Bernstein Polynomials (ditches $\varepsilon \rightarrow$ very hard)
- Ao Wang (JE 2022): use polynomial sieves: $\mathbb{E} \left[\left(\sigma_j^{-1}(\mathbf{s}_t; \mathbf{x}_t^{(2)}, F) - X_t^{(1)} \beta^{(1)} \right) \phi_k(Z_{jt}) \right] = 0$
- Lu, Shi, Tao (JE 2023): use partially linear model: $\log(s_{jt}/s_{0t}) = X'_{1,jt} \beta^0 + \psi^0(X_{2,jt}; IV_{J,t}) + \xi_{jt}$

$$\text{where } \psi^0(x_{2,jt}; IV_{J,t}) = \log \left[\frac{\int \frac{\exp(x'_{2,jt} v)}{\exp(IV_{J,t}(v))} f^0(v) dv}{\int \frac{1}{\exp(IV_{J,t}(v))} f^0(v) dv} \right].$$

But these are still **only as good as characteristics**.

Idea #1: t-STE Embeddings (Van Der Maaten and Weinberger, 2012)

- **Goal:** Assign a low-dimensional vector of characteristics $x_j \in \mathbb{R}^m$ for each product using triplets of the form “ j is more similar to k than to ℓ ”.
- The “t” in t-STE comes from using a **Student-t kernel** to model distances—giving heavy tails and better separation of clusters.

$$\max_{\mathbf{x}} \sum_{(j,k,\ell) \in \mathcal{T}} \ln(\pi_{jkl}) \quad \text{where} \quad \pi_{jkl} = \frac{\left(1 + \frac{\|x_j - x_\ell\|^2}{\alpha}\right)^{-\frac{\alpha+1}{2}}}{\left(1 + \frac{\|x_j - x_k\|^2}{\alpha}\right)^{-\frac{\alpha+1}{2}} + \left(1 + \frac{\|x_j - x_\ell\|^2}{\alpha}\right)^{-\frac{\alpha+1}{2}}}$$

- This is basically maximum (log) likelihood for a binary outcome model (closer to ℓ than k).
 - Fit with canned routine (which does gradient descent).
 - α is a (somewhat arbitrary) tuning parameter.

Unobserved Characteristics: Magnolfi Maclure Sorensen (2023)

FIGURE 1: Sample survey page

Which two cereals on the right are most similar to FROOT LOOPS?

				
	☐	☐	☐	☐
				
	☐	☐	☐	☐

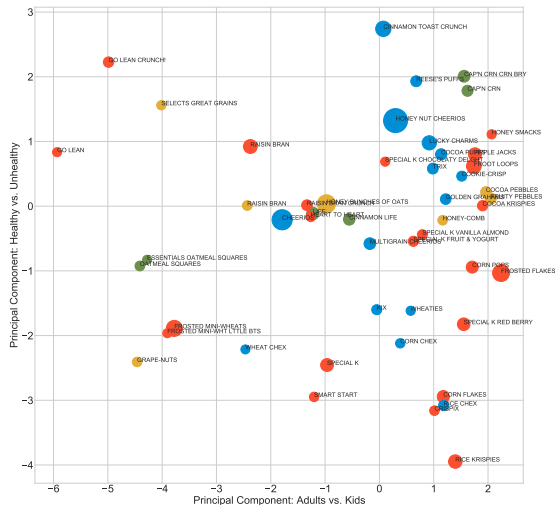
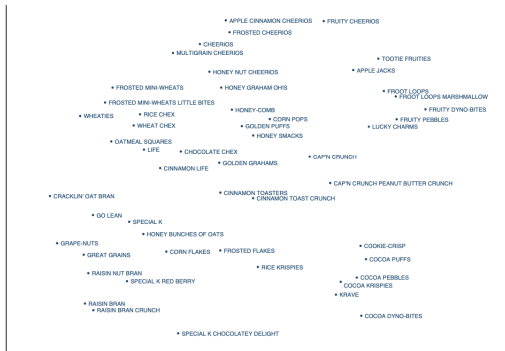
Submit

What if we could first estimate **unobserved characteristics**?

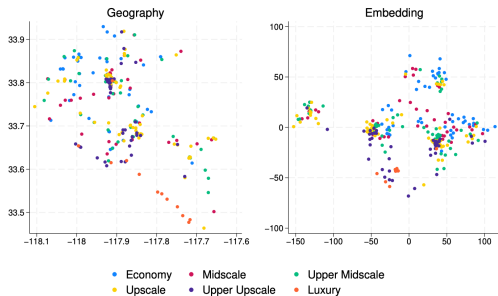
- Is j more similar to k or l ?
- Get a $m \times J$ matrix with m factors (embeddings).
- Idea: m is small (like 3-4).
- Use these as characteristics in BLP demand model.

Unobserved Characteristics: Magnolfi McClure Sorensen (2023)

FIGURE 2: Plot of two-dimensional embedding



Reverse Engineering Hotel Recommendations: McClure (2025)



APPENDIX FIGURE 1: Recommendations at Booking.com

Travelers who viewed Marriott Marquis Chicago ended up booking these properties

Show more



Hampton Inn Chicago McCor...

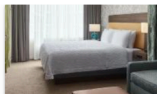
8.3 Very Good

📍 2 miles from center

♻️ Travel Sustainable property

Starting from

\$458



Home2 Suites By Hilton Chica...

8.4 Very Good

📍 2 miles from center

♻️ Travel Sustainable property

Starting from

\$441



Luxéry Stay Chicago - ACROS...

9.2 Wonderful

📍 1.9 miles from center

Starting from

\$500



Hilton Chicago

8.1 Very Good

📍 0.7 miles from center

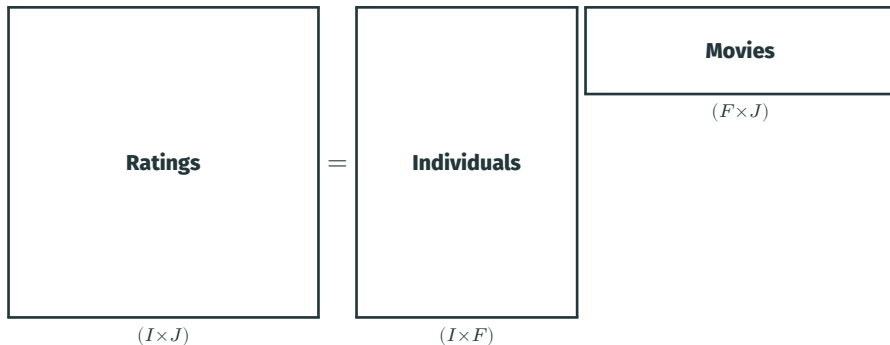
♻️ Travel Sustainable property

Starting from

\$544

- Scrape observed “you may also like” recommendations from hotels and use those to construct **triplets**
- Feed the triplets into tSTE embedding model to get a matrix $\mathbf{X}$: $(J \times F)$ where F is small (2-3).
- In both cases plug these in as characteristics to demand model (no guarantee they explain substitution patterns).

Idea #2: Low Rank Matrix Factorization: aka Netflix Prize



- ▶ Even if Ratings are **sparse**, we fit the observed cells and predict the rest!
- ▶ Idea: Approximate with a low rank (F) factor model (such as SVD)
- ▶ Maybe embeddings are reverse-engineering “collaborative filter”.

Second Choices as a Matrix: How Much Rank Do You Need?

For any mixture of logits, the second-choice (product-removal) diversion from j to k is

$$D_{j \rightarrow k} = \sum_{i=1}^I w_i \cdot \frac{\sigma_{ij}}{1 - \sigma_{ij}} \cdot \frac{\sigma_{ik}}{s_j} \iff \mathbf{D} = \text{diag}(\mathbf{s})^{-1} \tilde{\mathbf{S}} \text{diag}(\mathbf{w}) \mathbf{S}^\top, \quad \tilde{\sigma}_{ij} = \frac{\sigma_{ij}}{1 - \sigma_{ij}},$$

where the columns of $\mathbf{S}$ are the type-level choice-probability vectors σ_i .

Model	rank(D)	Why
Logit ($I = 1$)	≤ 1	one outer product $\tilde{\mathbf{s}} \mathbf{s}^\top$: proportional substitution
Nested logit (G nests)	$\leq G$	one outer product per (non-singleton) nest
Mixed logit (I types)	$\leq \min\{I, J + 1\}$	one rank-one term per type

- ▶ Logit and nested-logit bounds are **tight**; the mixed-logit bound is usually slack (thousands of draws on a handful of characteristics).
- ▶ Idea: skip characteristics and parametric random coefficients; **control rank(D) directly** in product space.

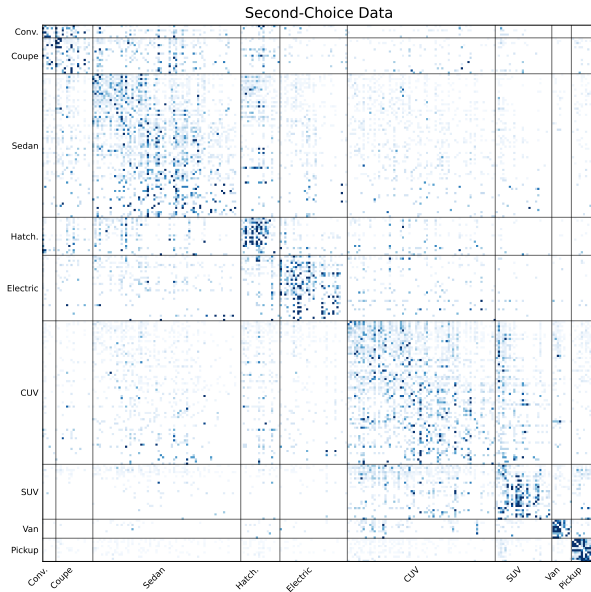
Conlon, Mortimer, Sarkis (2026): Second Choices in Product Space

Data from a **single market**: shares s_j and a subset Ω of survey second-choice cells $\mathcal{D}_{j \rightarrow k}$. For each rank I , solve

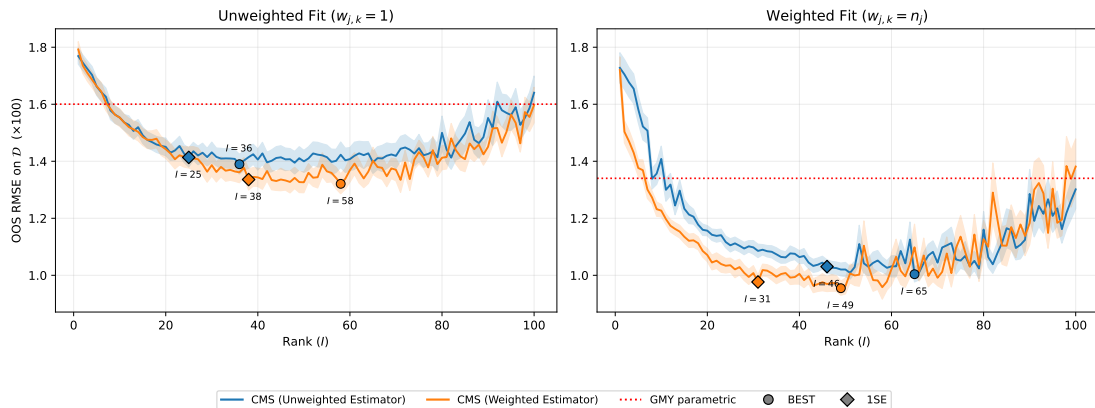
$$\min_{(\mathbf{S}, w) \geq 0} \sum_{(j,k) \in \Omega} \omega_{jk} (\mathcal{D}_{j \rightarrow k} - D_{j \rightarrow k}(\mathbf{S}, w))^2 + \lambda \|\mathbf{s} - \mathbf{S} w\|_{\ell_2}^2 \quad \text{s.t.} \quad \sum_j \sigma_{ij} = 1 \quad \forall i, \quad \sum_i w_i = 1.$$

- **Estimates** σ_i (type profiles) and w_i (type weights): a finite mixture with $I \ll J$ and no characteristics.
- **Tuning**: λ is a share-fit penalty (not a ridge); (I, λ) chosen by cross-validation on held-out cells of Ω .
- **Weights** $\omega_{jk} \propto n_j$, the respondents behind row j (approximately inverse-variance under row-multinomial sampling).
- **Computation**: reparameterize $\sigma_{ij} = e^{V_{ij}} / \sum_k e^{V_{ik}}, w_i = e^{\gamma_i} / \sum_\iota e^{\gamma_\iota}$, so the problem is unconstrained; autodiff + Adam. Optional sparsification zeroes $\geq 75\%$ of $\mathbf{S}$ (stop when AIC rises).
- **Matrix completion**: the fitted $\mathbf{D}(\mathbf{S}, w)$ predicts the *unobserved* cells $(j, k) \notin \Omega$.
- Software: [PyCMS](#) (PyTorch).

What the Data Look Like: MaritzCX Second Choices, 254 Cars and Light Trucks (2015)



Cross Validation: Out-of-Sample Fit



Held-out second-choice cells. Parametric BLP-style benchmark from Grieco, Murry, Yurukoglu (QJE 2023): $R^2 = 0.18$; CMS $I = 38$ (one-SE rule): 0.65; CMS $I = 58$ (best): 0.78.

Top Substitutes: Chevrolet Suburban (second-choice %)

Model	Raw	GMV (Parametric)	CMS 1SE (I=38)	CMS Best (I=58)
Gmc Yukon Xl	41.97	8.36	40.81	42.07
Ford Expedition	25.18	5.47	24.51	23.51
Toyota Sequoia	4.38	0.56	5.27	2.41
Chevrolet Tahoe	4.38	8.36	4.21	4.31
Cadillac Escalade	3.28	0.85	4.13	3.61
Gmc Yukon	2.19	2.99	2.83	4.42
Lincoln Navigator	1.82	0.69	5.73	5.15
Infiniti Qx80	1.46	0.48	0.47	0.74
Dodge Durango	1.09	2.20	0.73	0.79
Jeep Grand Cherokee	1.09	3.58	0.19	0.35

Raw survey vs. the parametric GMV model vs. CMS at two ranks. The parametric model spreads substitution across the segment; the data concentrate it on two near-twins.

Second Stage: Putting Prices Back In

The first stage delivers $\hat{\mathbf{S}}, \hat{w}$ and hence $V_{ij} - V_{i0}$, but no price sensitivity and no welfare. Assume $V_{ij} = V_{ij}^* - \alpha_i p_j$ (constant within type, free across types) and calibrate $\alpha_i > 0$ from one of:

- **Own-price elasticities** (a quasi-experiment, another study, a simpler demand system):

$$\min_{\alpha > 0} \left\| \mathcal{E}_{jj} + \frac{p_j}{\hat{s}_j} \sum_{i=1}^I \alpha_i \hat{w}_i \hat{\sigma}_{ij} (1 - \hat{\sigma}_{ij}) \right\|_{\ell_2} \quad (\text{need } \geq I \text{ targets, or } \alpha_i = \alpha)$$

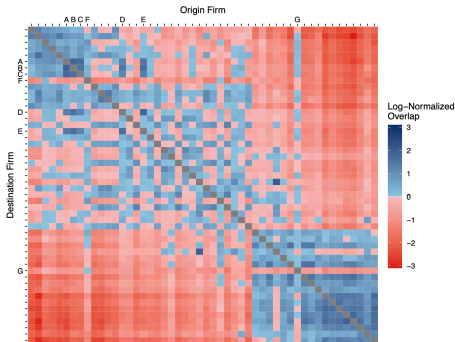
- **Price-cost margins:** Bertrand FOC with ownership matrix $\mathbf{H}$ and demand derivatives $\Delta(\alpha) = \sum_i \hat{w}_i \Delta_i(\alpha_i)$,

$$\mathbf{c}(\alpha) = \mathbf{p} - (\mathbf{H} \odot \Delta(\alpha))^{-1} \mathbf{s}, \quad \min_{\alpha > 0} \left\| \mathcal{C}_j - c_j(\alpha) \right\|_{\ell_2}.$$

- Either way you now have a full mixed logit in product space: elasticities, diversion under price changes, merger simulation (Week 7).
- **Inference:** each row of $\mathcal{D}$ is Multinomial($n_j, \cdot$); parametric bootstrap with warm-started re-solves. Type labels are not identified, so report CIs for **invariant functionals** only (fitted $D_{j \rightarrow k}$, elasticities). E.g. Suburban $\rightarrow$ Yukon XL + Expedition: 65%, 95% CI [60%, 71%].

Idea #3: Customer Overlap (Einav, Guido, Klenow 2025)

Figure 4: Hotel Overlap



Note: This picture visualizes all pairwise log-normalized overlap measures for the top 50 hotel chains in the U.S. Hotel chains sorted from highest non-hotel spending rank to lowest non-hotel spending rank across both axes, going from left to right and from top to bottom (i.e. highest ranked chains are in the top left and lowest ranked chains are in the bottom right). A single cell of this matrix can be interpreted as the log-normalized overlap of the column chain with the row chain (i.e. $C_{col \rightarrow row}$). Thus, any column can be interpreted as how much that chain's customers relatively overlap with other chains. Similarly, any row can be interpreted as how much all other chains' customers overlap with the row chain. Values above zero are colored blue, with darker colors indicating higher log-normalized overlap. Similarly, values below zero are colored red, with darker shades

- ▶ Encode a $J \times 1$ vector $\mathbf{q}_i$ of every purchase (0/1) within the category for the year
- ▶ They use credit card data (don't see items, only stores).
- ▶ Could use Nielsen panelists.
- ▶ How many customers do two stores have in common? $C_{j \rightarrow k} = \frac{\sum_{i \in \mathcal{C}_j} q_{i,k}}{\sum_{i \in \mathcal{C}_j} q_{i,(-j)}}$.
- ▶ Use to estimate diversion ratios (but not a demand model).
- ▶ Conlon Rao (JPE, forthcoming) use a similar measure to estimate nesting parameter.
- ▶ Atalay et. al (JPE, 2024) use a similar measure to assign products to nests.

Thanks!
